Introduction to Relational Databases

A database stores records in tables, and every table has a primary key that identifies each record uniquely. A foreign key links a record in one table to a record in another table.

Normalization removes redundancy from a relational database design by splitting data into well structured tables. Designers draw an ER diagram of each entity before creating the schema in the DBMS.

SQL is the language used to query a relational database. The SELECT statement retrieves records, and a JOIN combines records from several tables using their keys.

A transaction groups several SQL operations so that they either all succeed or all fail, which keeps the information system consistent.